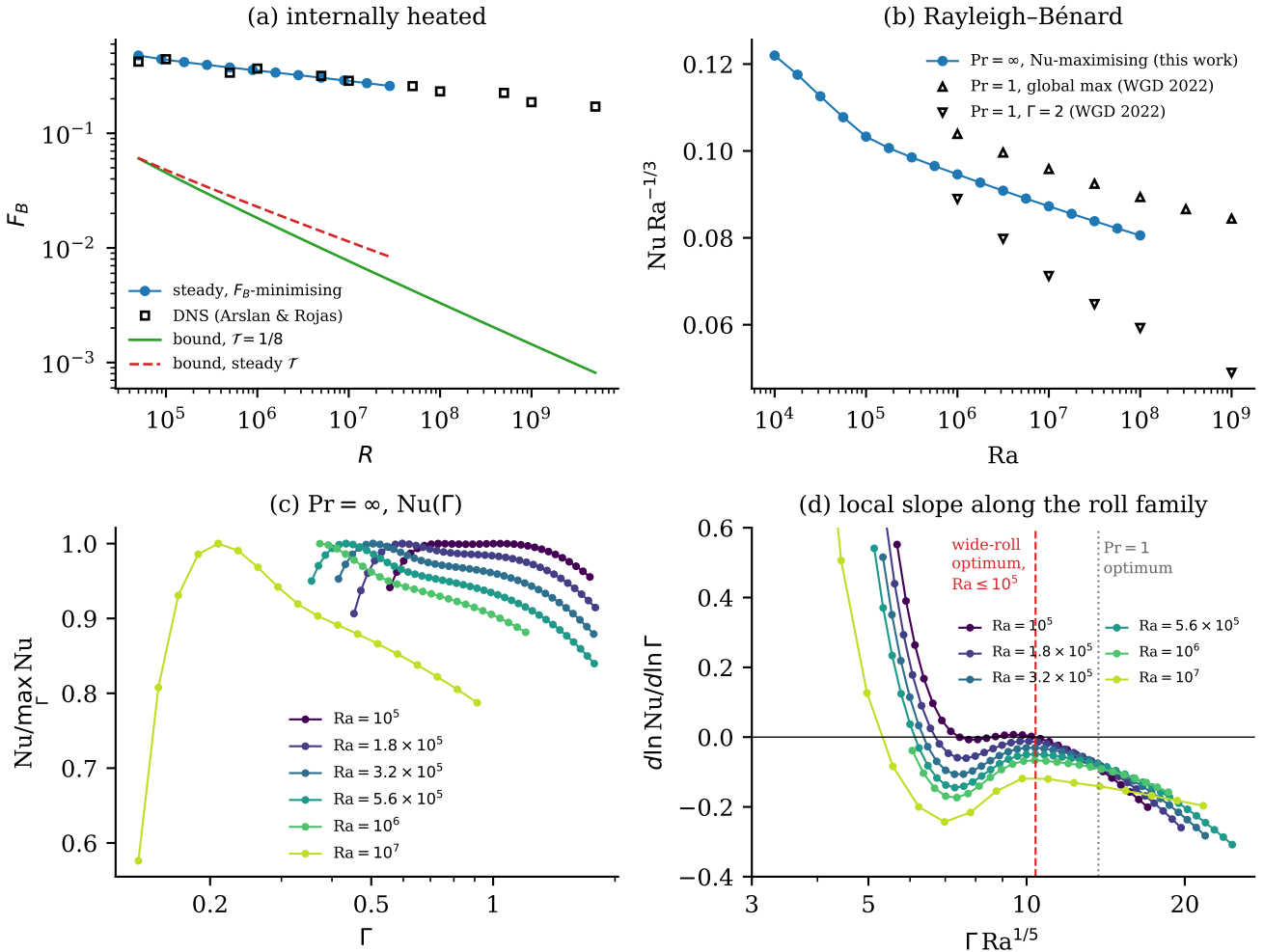
(a) internally heated
F_B
R
steady, F_B-minimising
DNS (Arslan & Rojas)
bound, $\mathcal{T}=1/8$
bound, steady $\mathcal{T}$
(b) Rayleigh–Bénard
$\mathrm{Nu}\,\mathrm{Ra}^{-1/3}$
Ra
Pr = ∞, Nu-maximising (this work)
Pr = 1, global max (WGD 2022)
Pr = 1, Γ = 2 (WGD 2022)
(c) Pr = ∞, Nu(Γ)
$\mathrm{Nu}/\max_\Gamma \mathrm{Nu}$
Γ
Ra = 10^5
Ra = 1.8×10^5
Ra = 3.2×10^5
Ra = 5.6×10^5
Ra = 10^6
Ra = 10^7
(d) local slope along the roll family
$d\ln\mathrm{Nu}/d\ln\Gamma$
$\Gamma\,\mathrm{Ra}^{1/5}$
wide-roll optimum, Ra ≤ 10^5
Pr = 1 optimum